

Homework 8: Guided QAOA/QML Paper Report

Quantum Computing Training 2026

Choose one research paper about quantum approximate optimization algorithms, variational quantum algorithms, or quantum machine learning. In this assignment, you are asked to write a ~2 page report summarizing the key ideas, limitations, and possible extensions of the paper you choose. You may choose a paper from the lecture, from arXiv, from a journal, or from another reliable source. If you are unsure whether a paper is appropriate, ask me.

The paper you choose should be technical enough that there is a real algorithm, model, experiment, theorem, or benchmark to explain. Avoid papers that are purely promotional or purely philosophical. Good choices include papers about:

- QAOA for MaxCut, constraint satisfaction, scheduling, routing, portfolio optimization, etc.;
- Generalized QAOA mixers or constrained optimization;
- QAOA performance, trainability, or benchmarking against classical methods;
- Quantum kernels, variational quantum classifiers, quantum neural networks, or quantum generative models;
- Limitations of QAOA or QML, such as barren plateaus, data loading, shot noise, noise sensitivity, or weak classical baselines.

Assume your audience is the set of students in this course. Ideally, I would like to post your report to the course's Github repository so that the other students in the course can read it.

Part A: Core summary

- A.1. Paper Choice.** What paper have you chosen? The title, the authors, the journal/arXiv number, and the date? Is this a QAOA or a QML paper?
- A.2. Problem setting.** What computational problem, learning task, or optimization problem is being studied? What are the inputs and outputs?
- A.3. Quantum circuit or model.** What quantum circuit, ansatz, feature map, mixer, Hamiltonian, or measurement is used? If the paper uses QAOA, identify the cost objective $C(z)$ or cost Hamiltonian H_C and the mixer. If the paper uses QML, identify the data encoding, trainable parameters, and measured observable or loss.
- A.4. Classical outer loop.** What is optimized classically? Examples include QAOA angles (γ, β) , variational circuit parameters θ , kernel hyperparameters, or model weights.
- A.5. Evidence.** What evidence does the paper provide? For example, does it give a theorem, a numerical experiment, a hardware experiment, a comparison to classical baselines, or a scaling argument?
- A.6. Main result.** State the paper's main result in your own words. Be precise.

Part B: Critical analysis

- B.1.** What assumptions does the paper make? Examples: noiseless circuits, access to many shots, efficient state preparation, small datasets, special graph families, or a particular optimizer.
- B.2.** What is the strongest part of the paper? This could be a clean idea, a useful benchmark, a clever circuit construction, a convincing comparison, or a theorem.
- B.3.** What is one limitation or weakness? For example, is the dataset too small, the classical baseline weak, the circuit too deep, the optimization unstable, or the resource accounting incomplete?
- B.4.** If the paper includes experiments or simulations, what would you want to reproduce first? Define the smallest reproducible piece of the paper. Try reproducing it.

Part C: Proposed extension or project idea

The last part of your report should propose a possible next step inspired by the paper. Your proposal should be **well-motivated, well-defined, and testable**. It does not need to be guaranteed to work. It should be something that a student or small team could begin investigating.

- C.1.** What gap, limitation, unanswered question, or extension of the paper motivates your idea?
- C.2.** Briefly state your proposed research question.
- C.3.** Give a concrete project plan. Your plan should say what you would implement, what baseline(s) you would compare against, and what metric(s) you would report.
- C.4.** What result would support your idea? What result would falsify it or make it less promising?
- C.5.** What would you need to carry this out? Examples: Qiskit simulation, a specific dataset, graph instances, a classical optimizer, access to hardware, or more background reading.

You do **not** need to implement code for this homework. However, your proposed extension should be specific enough that someone could start implementing it (if needed) after reading your report.

Deliverables

Submit a PDF containing:

- The paper citation;
- A ~ 2 page report following Parts A–C;
- Any useful figures, tables, or diagrams;
- A short references section.